MD HASANUR RAHMAN

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EDUCATION

University of Florida *2025–Present*
PhD Candidate in Electrical and Computer Engineering
Advisor: Guanpeng Li

University of Iowa *2021–2025*
MS in Computer Science, 2024
PhD studies in Computer Science prior to transfer to University of Florida
Advisor: Guanpeng Li

Bangladesh University of Engineering and Technology *2015–2019*
BS in Computer Science and Engineering

RESEARCH INTERESTS

High Performance Computing, Dependable Systems, Data Reduction, Parallel Computing, Program Analysis, Efficient and Reliable LLM Inference

SELECTED RESEARCH CONTRIBUTIONS

- **Conservative Silent Error Risk Assessment:** Developed input-generation methods that identify high-risk application executions and enable conservative evaluation of silent data corruption (SDC) vulnerability.
- **Input-Aware Adaptive Protection:** Developed adaptive lightweight protection techniques, improving reliability by protecting regions that are more likely to produce SDC errors.
- **Data-Aware Compression Decision Support:** Developed intelligent frameworks that help users efficiently that guide compression choices under storage, data-transfer, and quality constraints.

PUBLICATIONS

Peer-Reviewed Conference Publications

- **Temporal Modeling of Soft Error Propagation Using Dynamic Bayesian Networks**
  Ahmer Jamil, Md Hasanur Rahman, Bingzhe Li, Xiaoyi Lu, Guanpeng Li
  IEEE International Conference for High-Performance Computing, Networking, Storage and Analysis (SC), 2026. To appear.
  Acceptance rate: 19.0%
- **Deploying Lightweight Input-Aware Selective Instruction Duplication in HPC Applications**
  Md Hasanur Rahman, Guanpeng Li
  ACM International Conference for High-Performance Computing, Networking, Storage and Analysis (SC), 2025
  Acceptance rate: 21.2%
- **Modeling Rate-Distortion for Endpoint-Aware Lossy Compression in Scientific Data Transfer**
  Md Hasanur Rahman, Sheng Di, Guanpeng Li, Franck Cappello
  IEEE International Performance Computing and Communications Conference (IPCCC), 2025
  Acceptance rate: 32.1%

- **DRUTO: Upper-Bounding Silent Data Corruption Vulnerability in GPU Applications**
  Md Hasanur Rahman, Sheng Di, Shengjian Guo, Xiaoyi Lu, Guanpeng Li, Franck Cappello
  IEEE International Parallel & Distributed Processing Symposium (IPDPS), 2024
  Acceptance rate: 25.0%
- **A Generic and Efficient Framework for Estimating Lossy Compressibility of Scientific Data**
  Md Hasanur Rahman, Sheng Di, Guanpeng Li, Franck Cappello
  IEEE International Conference on Massive Storage Systems and Technology (MSST), 2024
- **Significantly Improving Fixed-Ratio Compression Framework for Resource-Limited Applications**
  Tri Nguyen, Md Hasanur Rahman, Sheng Di, Michela Becchi
  International Conference on Parallel Processing (ICPP), 2024
  Acceptance rate: 29.0%
- **A Feature-Driven Fixed-Ratio Lossy Compression Framework for Real-World Scientific Datasets**
  Md Hasanur Rahman, Sheng Di, Kai Zhao, Robert Underwood, Guanpeng Li, Franck Cappello
  IEEE International Conference on Data Engineering (ICDE), 2023
  Acceptance rate: 19.1%
- **Characterizing Deep Learning Neural Network Failures between Algorithmic Inaccuracy and Transient Hardware Faults**
  Sabuj Laskar, Md Hasanur Rahman, Bohan Zhang, Guanpeng Li
  IEEE Pacific Rim International Symposium on Dependable Computing (PRDC), 2022
  Acceptance rate: 36.0%
- **Peppa-X: Finding Program Test Inputs to Bound Silent Data Corruption Vulnerability in HPC Applications**
  Md Hasanur Rahman, Aabid Shamji, Shengjian Guo, Guanpeng Li
  ACM International Conference for High-Performance Computing, Networking, Storage and Analysis (SC), 2021
  Acceptance rate: 23.6%

Peer-Reviewed Journal Publications

- **A Survey on Error-Bounded Lossy Compression for Scientific Datasets**
  Sheng Di, Jinyang Liu, Kai Zhao, Xin Liang, Robert Underwood, Zhaorui Zhang, Milan Shah, Yafan Huang, Jiajun Huang, Xiaodong Yu, Congrong Ren, Hanqi Guo, Grant Wilkins, Dingwen Tao, Jiannan Tian, Sian Jin, Zizhe Jian, Daoce Wang, Md Hasanur Rahman, Boyuan Zhang, Shihui Song, Jon Calhoun, Guanpeng Li, Kazutomo Yoshii, Khalid Alharthi, Franck Cappello
  ACM Computing Surveys, 2025
- **Investigating the Impact of Transient Hardware Faults on Deep Learning Neural Network Inference**
  Md Hasanur Rahman, Sabuj Laskar, Guanpeng Li
  Journal of Software Testing, Verification and Reliability (STVR), 2024

Workshop Publications

- **Characterizing Spatial Data Traits for Modeling Generic Lossy Rate-Distortion Quality**
  Md Hasanur Rahman, Sheng Di, Guanpeng Li, Franck Cappello
  IEEE International Parallel & Distributed Processing Symposium Workshops (IPDPS-W), 2025
- **LibPressio-Predict: Flexible and Fast Infrastructure for Inferring Compression Performance**
  Robert R. Underwood, Sheng Di, Sian Jin, Md Hasanur Rahman, Arham Khan, Franck Cappello
  ACM International Workshop on Data Reduction for Big Scientific Data (DRBSD), in conjunction with SC, 2023

- **TensorFI+: A Scalable Fault Injection Framework for Modern Deep Learning Neural Networks**
  Sabuj Laskar, Md Hasanur Rahman, Guanpeng Li
  IEEE International Workshop on Resiliency, Security, Defences and Attacks (ISSRE-W), 2022

INDUSTRY EXPERIENCE

Samsung Research *2019–2021*
Software Engineer *Bangladesh*

HONORS AND AWARDS

- SC Travel Grant, 2026.
- Graduate Research Excellence Award Finalist, 2025.
- SC Travel Grant, 2024.
- ICDE Travel Grant, 2023.

PROFESSIONAL SERVICE

- **Technical Program Committee:** IEEE CLOUD Summit, 2026.
- **Artifact Evaluation Committee:** IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), 2024.
- **Subreviewer:** DSN 2026; HiPC 2025; ISSRE 2024, 2023, 2022; HPDC 2023, 2022; DSN 2023, 2022; Middleware 2022; SELSE 2022; PRDC 2021.
- **Student Volunteer:** SC 2024; ICDE 2023.

STUDENT MENTORING

- Ahmer Jamil, 2025–2026.
- Abdullah Naveed, 2024–2026.
- Sabuj Laskar, 2022–2023.